

Research paper

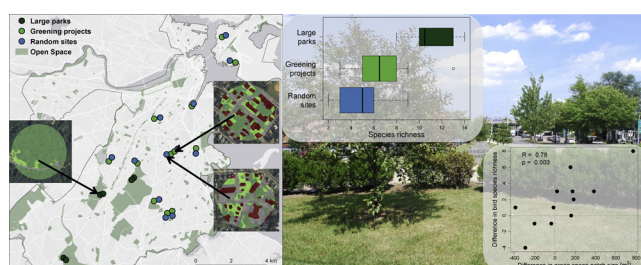
Are small greening areas enhancing bird diversity? Insights from community-driven greening projects in Boston

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HIGHLIGHTS

- ▶ We studied the effect of small greening projects on birds in Boston, MA.
- ▶ Greening projects have slightly more species than random sites in their vicinity.
- ▶ Average patch size explains most of this difference.
- ▶ Our results indicate tradeoffs with environmental justice criteria for greening.

GRAPHICAL ABSTRACT



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ABSTRACT

Green space is an important component of the urban landscape, providing ecosystem services for city dwellers and supporting biodiversity. In many cities, green space is concentrated in large areas, while the rest is dispersed in small patches like pocket parks, gardens or street trees. Such small-scale green space is often the target of greening initiatives but little is known about their value for supporting and conserving biodiversity. To assess the value of such initiatives for biodiversity, we used birds as an indicator. We examined bird communities at small greening projects ($n = 12$), nearby randomly chosen urban sites ($n = 12$), and large parks ($n = 6$) in Boston, MA, in relationship to underlying landscape patterns. Our results show that large parks harbor a distinct and rich bird community. The other sites, regardless of the presence of greening projects, were very similar to each other. However, most sites with greening projects had higher species richness than the random urban site in their vicinity. The main factor associated with this appears to be the patch size of green space and to a lesser extent, tree cavities. Even small increases of a few hundred square meters were associated with an increase in bird richness. Having more trees with cavities was also beneficial for species richness. Small greening projects appear to be most valuable for urban biodiversity if they target preserving, increasing and connecting existing green space. This may represent a tradeoff with environmental justice criteria for new green space.

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1. Introduction

Protecting and supporting urban biodiversity is a growing concern for conservationists because urbanization alters entire landscapes, drastically impacting wildlife, with many species groups, in particular birds, responding negatively to urbanization (Grimm et al., 2008; Kareiva, Watts, McDonald, & Boucher, 2007). Species diversity and functional diversity of birds is much lower in densely built places than in rural areas (Chace & Walsh, 2006). Along rural-urban gradients, bird richness and diversity declines

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with decreasing vegetation cover (Blair, 2004; Donnelly & Marzluff, 2006; MacGregor-Fors & Schondube, 2011) and distance to the urban fringe (MacGregor-Fors & Ortega-Álvarez, 2011). Sometimes, this relationship is hump-shaped with highest diversity at intermediate urbanization levels (Blair, 1996; Tratalos et al., 2007). Within cities the urban form (e.g., housing density and building types; Pellissier, Cohen, Boulay, & Clergeau, 2012), and vegetation composition and structure (Fontana, Sattler, Bontadina, & Moretti, 2011; Lerman & Warren, 2011; White, Antos, Fitzsimons, & Palmer, 2005) has been shown to influence species composition and diversity. Urbanization also changes functional aspects of the bird community and has been shown to cause a loss of specialist guilds (Clergeau, Croci, Jokimäki, Kaisanlahti-Jokimäki, & Dinetti, 2006; Devictor, Julliard, Couvet, Lee, & Jiguet, 2006). In particular, insectivorous species have been shown to decline with increasing urbanization (Chace & Walsh, 2006). By contrast, the relative abundance of omnivores and granivores often increases with urbanization (Chace & Walsh, 2006), although there are exceptions. For example, MacGregor-Fors (2008) found more insectivores than granivores except at the highest urbanization level. In some cities, like Phoenix, AZ, USA, Vancouver, Canada, and Leipzig, Germany, species richness has been found to be related to socioeconomic status (Kinzig, Warren, Martin, Hope, & Katti, 2005; Melles, 2005; Strohbach, Haase, & Kabisch, 2009).

Highly urbanized areas, independent of socioeconomic status, are often species poor and dominated by generalist and introduced species (Bolger, Scott, & Rotenberry, 1997; Cohen, Baudoin, Palibrk, Persyn, & Rhein, 2012; Loss, Ruiz, & Brawn, 2009; Melles, Glenn, & Martin, 2003; White et al., 2005). However, cities are not entirely hostile to birds. In particular, large green spaces within the city, e.g., parks, cemeteries, golf-courses, wetlands, and forests, play an important role for urban biodiversity and can support species rich bird communities (Blair, 2004; Donnelly & Marzluff, 2004; Sandström, Angelstam, & Mikusiński, 2006). Patch size is one of the most important factors for explaining bird diversity in green space, as predicted by the theory of island biogeography (Fernández-Juricic & Jokimäki, 2001). Donnelly and Marzluff (2004) estimated that 42 ha is a threshold size that allows native forest bird species to persist in green space patches in the Seattle metropolitan area and Fernández-Juricic and Jokimäki (2001) found that parks between 10 and 35 ha in size contain most bird species that are found in the overall city area.

Boston, Massachusetts, USA, is a city with both high-density urban development and a large, interconnected green way (Urban Ecology Institute, 2008). It is one of the oldest European settlements in New England, founded in 1630. The city and the surrounding towns grew rapidly in the 19th century (Kennedy, 1994) and the changes in the bird community documented for a small residential area near Boston give an impression about the scale of the impacts of urbanization. Between 1860 and 1964, the number of nesting species found in one neighborhood declined from 26 to 9 (Walcott, 1974). The author attributes these changes to several factors, most important, the loss of open and undeveloped space. During this time of rapid growth, the city's park system was established based on designs by Frederic Law Olmsted, preserving a string of interconnected larger parks from development (Kennedy, 1994). The Emerald Necklace, as it is called, still contains half of the city's parkland (City of Boston, n.d.), and its conservation value is highlighted by Aversa (1994), who reported 134 bird species in a migration census in Franklin Park, the largest park of the greenway.

As in other cities, green space in Boston is not equally distributed among the human population (Urban Ecology Institute, 2008). In awareness of the importance of green space for human well-being and biodiversity, the city of Boston has stated the ambitious goal of increasing the size of its urban forest area from 29% to 35% of the city area by 2030 (Boston Natural Areas Network, n.d.).

To reach this goal, the "Grow Boston Greener" program provides non-profit organizations with mini-grants for tree plantings. "Grow Boston Greener" aims especially on underserved parts of the city (Boston Natural Areas Network, n.d.). A similar community-driven approach to greening in the Boston area is the "CityRoots" initiative (<http://www.urbaneco.org/city-roots.html>). The spatial scale of such projects is small (usually below 0.5 ha), because they are carried out by volunteers from the local community and are implemented in residential areas that are already densely built.

Little is known about the value of such small projects for supporting and conserving local biodiversity but it is a timely issue considering the high competition for space within densely built and growing cities (Carbó-Ramírez & Zuria, 2011; Fernández-Juricic & Jokimäki, 2001). Fuller and Gaston (2009) estimate that interactions between people and nature will increasingly depend on non-formal green space such as street plantings. Small green spaces are thought to be less valuable for birds because they have more edge habitat, less habitat diversity, provide less cover from predators and human disturbance than larger parks, and they support smaller populations and generally less individuals (Donnelly & Marzluff, 2004; Evans, Newson, & Gaston, 2009; Fernández-Juricic & Jokimäki, 2001; Harrison, Burgess, Millward, & Dawe, 1995). However, other local factors such as tree age, vegetation structure and composition, spatial context of green space and connectivity with other habitat, human disturbance, and supplementary feeding are also influential (Evans et al., 2009; Fernández-Juricic & Jokimäki, 2001; Fontana et al., 2011). Such local factors have been shown to be more important than regional ones, indicating a potential for small-scale greening efforts to have a positive influence on urban biodiversity (Evans et al., 2009).

In this paper, we explored the value of community-driven greening projects for supporting and conserving local avian diversity in Boston, MA. We studied bird diversity and community composition at greening projects, in order to gain a better understanding of whether they show detectable differences compared to non-greened urban areas in their vicinity, and why this may be the case. Because we were interested in local differences, we paired small greening projects with adjacent randomly selected urban sites. For comparison, large wooded parks, representing more natural habitat, were also surveyed. We examined differences in the bird community and species richness between the greening projects and random sites as well as the large parks. In addition, we assessed which characteristics of the sites and their surrounding explained potential patterns. Finally, we discuss the relevance of our findings for bird conservation in cities.

2. Methods

2.1. Study area and sampling design

Boston is located at the Massachusetts Bay in the northeastern USA (42°21'N, 71°4'W). The climate is classified as snow climate, fully humid with hot summers (Kottek, Grieser, Beck, Rudolf, & Rubel, 2006). The city has 617,594 inhabitants and a population density of 4939 inhabitants per km² (United States Census Bureau, n.d.). The entire metropolitan area is ca. 3731 km² in size and home to 6,547,629 people (Metropolitan Area Planning Council, n.d.; Fig. 1). The "Emerald Necklace" park system encompasses 445 ha, which is 50% of the city's parkland (City of Boston, n.d.). Tree canopy covers ca. 29% of the city area but it is not evenly distributed, ranging between 6% and 49% in the municipal districts (Urban Ecology Institute, 2008).

We designed our study in conjunction with the "CityRoots" greening campaign by the Urban Ecology Institute, which aims at greening in underserved neighborhoods through community

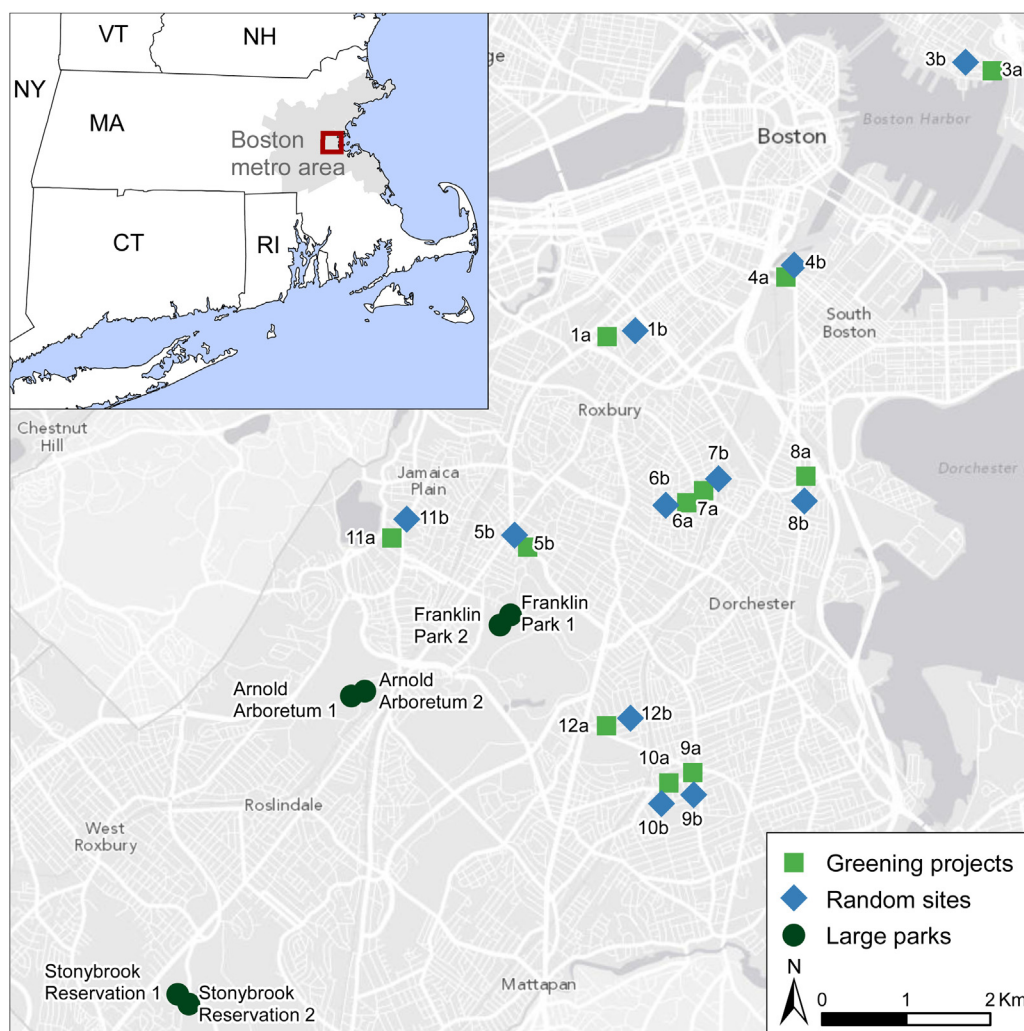


Fig. 1. Overview map of the 30 bird observation sites in Boston, Massachusetts. The IDs refer to Table 1 and Figs. 2 and 3. Base map © Esri, DeLorme, NAVTEQ.

planting projects. In seven years, 48 “CityRoots” projects were implemented involving more than 600 residents, ranging in scale from street tree plantings to the creation of green space on vacant lots (Urban Ecology Institute, n.d.). Because the “CityRoots” program aims at underserved communities, they are mostly implemented in areas with little existing green space. Not all 48 “CityRoots” could be sampled, nor were they suitable. We selected projects from within the city of Boston that involved tree plantings because the city is embedded in a forested ecosystem. Thus, we expected that tree planting would have the greatest impact on

birds. Most “CityRoots” projects were carried out in less affluent neighborhoods. Because we wanted to minimize the influence of socioeconomic status, a factor influential in other studies (Kinzig et al., 2005; Melles, 2005; Strohbach et al., 2009), we restricted our study to projects located in medium and low-income census block groups (2000 US census, <\$50,000 median household income). As a consequence of being carried out by neighborhood associations, the “CityRoots” projects tended to be clustered in certain areas. Therefore, we selected projects that were spread throughout the city. Our final selection included 12 greening projects. The sites, like the

Table 1

Description of the 12 “CityRoots” greening projects. The map ID refers to Figs. 1–3. The extent of the sites is an approximation. If the greening aimed at enhancing and maintaining an existing site, the extent of the original site is shown.

Map ID	Description	Year of greening	Green space extent, new or upgraded (m ²)
1a	Creation of small park with trees and shrubs	2005–2007	424
2a	Planting of trees on school yard	2003	52
3a	Tree plantings in existing park	2005–2006	4377
4a	Planting of street trees	2008	108
5a	Creation of ornamental community garden	2009	973
6a	Greening of vacant lot with trees, shrubs and perennials	2005	251
7a	Tree plantings at residential houses	2008	859
8a	Tree plantings in park	2006–2008	1290
9a	Greening in small park with trees, shrubs and perennials	2008	369
10a	Greening in small park with trees, shrubs and perennials	2009	271
11a	Greening at residential houses with trees and perennials	2007	1319
12a	Tree planting in ornamental community garden	2005–2006	615

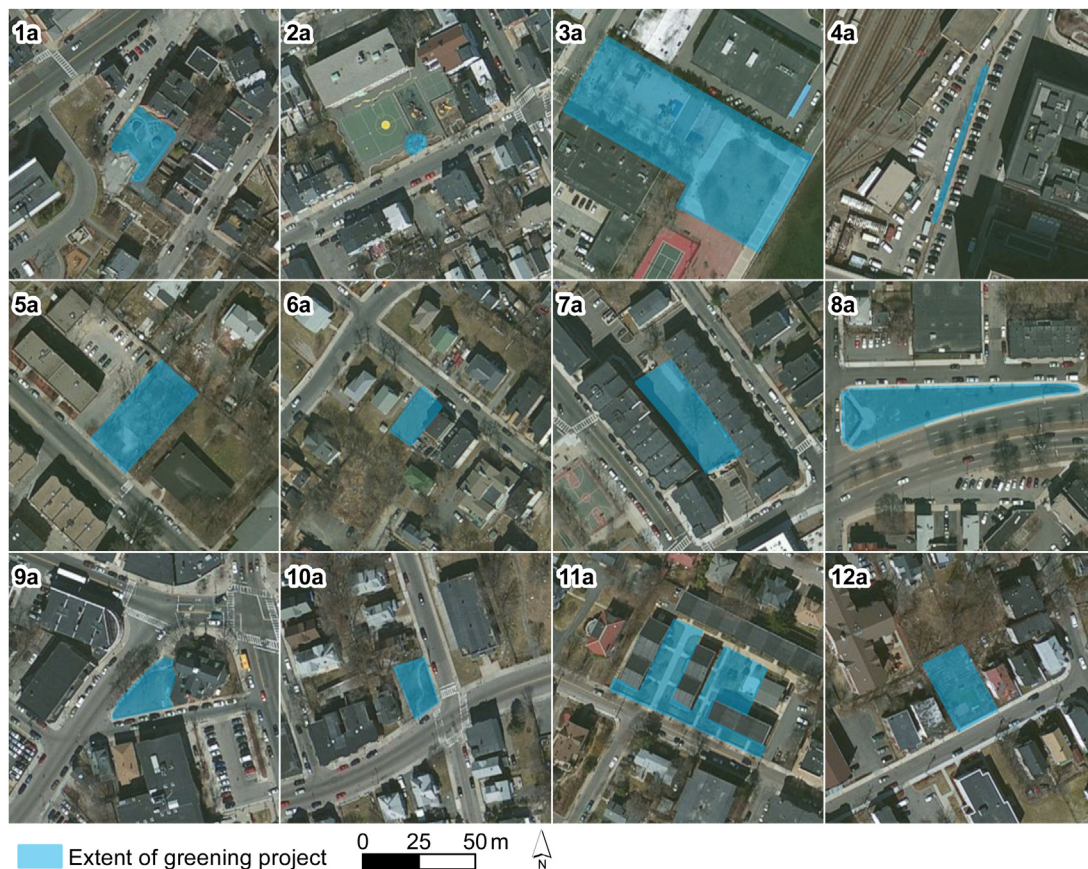


Fig. 2. Maps of the estimated extent of the 12 “CityRoots” projects in Boston, Massachusetts. If the greening aimed at enhancing and maintaining an existing site, the extent of the original site is shown. The IDs refer to Table 1 and Figs. 1 and 3. Base map © Esri, DeLorme, NAVTEQ.

“CityRoots” projects in general, are very diverse. Two projects created entirely new green spaces, one involved greening a vacant lot, five aimed at improving and maintaining existing small parks, and four involved greening at schoolyards, roads or residential houses. The average size was approximately 900 m². With one exception (2003), the projects were carried out between 2005 and 2009 (Figs. 1 and 2 and Table 1).

One bird observation point was placed at each greening project. In order to test whether different bird communities and species richness could be found compared to the surrounding urban matrix, each greening project was paired with a randomly located site nearby. Because the sites had to be accessible, we constrained the random site selection to public roads 250–300 m apart from the greening projects and then moved the observation points manually to the nearest sidewalk. The distance of 250 m was chosen to ensure that the same birds were not counted twice, as recommended by Ralph, Droege, and Sauer (1995). In addition, we sampled three large parks: Arnold Arboretum (110 ha) and Franklin Park (210 ha), both belonging to the Emerald Necklace, and Stony Brook Reservation (190 ha). One sampling location was randomly assigned for each large park and paired with a second plot 250 m apart from the first to match the sampling design for the greening project sites (Fig. 1).

2.2. Bird community

We used unlimited radius point counts to sample birds at the greening sites, associated random sites, and the large parks. Unlimited distance point counts are the most widely used counting technique for birds and they are particularly suitable to detect population responses to habitat differences (Ralph et al., 1995). All

birds seen or heard within a period of 5 min from a fixed point were recorded along with the estimated distance to the observer. Birds only seen in flight were recorded but omitted from further analysis because it is not clear whether these birds interacted with the habitat at the study sites. Owing to the fact that we compared the greening projects and random sites with large parks that were forested, we omitted birds observed farther than 50 m from the observer from further analysis because 50 m is the most practical radius used for point counts in forested areas (Ralph et al., 1995). All point counts were conducted in the breeding season between 6 am and 10 am in early June (1st–12th) and again in late June (18th–30th) of 2010 and 2011. In total, each site was visited four times for the study with the exception of sites 8b and 11b that could only be visited three times.

Volunteer birders from the Boston area conducted the sampling as part of a citizen science project. They were provided with maps and a detailed birding instruction and were expected to familiarize themselves with estimating distances but were allowed to decide independently about when to conduct their counts within the constraints described above. They were not told which sites were greening projects. To keep the time commitment for the volunteers to a minimum, we did not test for their ability to detect birds. False positives are unlikely: we checked the species lists for plausibility and birders were very cautious in marking observations when they were not certain. To account for false negatives (missed species), we took measures recommended by McCaffrey (2005). First, we assigned different birders to the early and the late June observation and varied observers between the two years. The data from the four observations were then combined to create a species list for each site. A total of nine birders volunteered for our study over the two years. Species accumulation curves (see Section 3)

show that saturation was achieved for the greening projects and random urban sites but not for the large parks. We considered this acceptable since the large parks were points of reference while our main focus was the relative differences between the small greening projects and the random urban sites.

2.3. Environmental parameters

Because we expected that differences between greening projects and random urban sites might arise due to characteristics of the greening projects and their direct surrounding, we measured a set of potential environmental covariates. We selected parameters found in previous studies to influence birds in urban environments: land cover, canopy cover and tree structure (Blair, 2004; Chace & Walsh, 2006; Donnelly & Marzluff, 2006; MacGregor-Fors & Ortega-Álvarez, 2011), distance to existing green space (Donnelly & Marzluff, 2004) and land-use/socioeconomic parameters (Cohen et al., 2012; Kinzig et al., 2005; Lerman & Warren, 2011; Loss et al., 2009; Melles, 2005; Strohbach et al., 2009).

We digitized the current land cover based on the 2010 National Agriculture Imagery Program (NAIP) orthophotos in a 100 m buffer, a size found to be relevant by Hostetler and Knowles-Yanez (2003) for several species and chosen for studying local conditions by Bolger et al. (1997) and Carbó-Ramírez and Zuria (2011). The following land-cover classes were differentiated: bare soil, building, lawn, impervious surface, tree and water. We used the percentage cover and average patch size of each land cover class as variables. We also combined adjacent lawns and trees to estimate the total cover and average patch size of green space. As a measure of tree structure and tree maintenance level, we conducted a tree survey along a 100 m transect that was centered at each bird observation point. At the greening projects and the random urban sites, the transects ran parallel to the road but included trees in green space and front yards. Ten randomly selected trees within or near a transect were identified and measured. Measurements included height, diameter at breast height (DBH), and the presence of cavities. The distance to green space larger than 42 ha was calculated in ArcGIS, using the “Protected and Recreational OpenSpace” layer (Commonwealth of Massachusetts, n.d.), based on the importance of reserve size (Donnelly & Marzluff, 2004). Finally, as a socioeconomic parameter, we selected all property parcels from the 2010 US census that intersected with a buffer radius of 100 m around the bird counts and extracted information on parcel size, parcel value and parcel development age. All 24 parameters are listed in Table 2.

To study how functional aspects of the bird community vary among the sites, foraging guilds for each species were extracted from DeGraaf and Wentworth (1986). For birds with multiple foraging guilds, we used the guild assigned for the breeding season. Substrate types used for foraging were also extracted.

2.4. Statistical analysis

All statistical analyses were carried out in the R statistical environment using functions provided with the packages *stats*, *ecodist* and *vegan* (Goslee & Urban, 2007; Oksanen et al., 2012; R Development Core Team, 2010). For the statistical analysis, we combined the 2010 and 2011 bird counts and removed all entries that were only determined to genus level. We also removed shore birds from the list, because this group was only present at some sites close to the coast.

2.4.1. Bird community

To explore patterns in the bird community and their relationships to environmental data, we conducted a nonmetric multidimensional scaling (NMDS) ordination. A NMDS is an

Table 2

Results from fitting environmental parameters on the ordination seen in Fig. 3. The correlation coefficient R^2 and the significance p are provided. Significance was calculated from 1000 permutations.

Parameter	R^2	p	Direction on ordination in Fig. 3
Bare soil cover	.22	.	←
Building cover	.53	***	→
Lawn cover	.25		
Impervious surface cover	.78	***	→
Tree canopy cover	.84	***	←
Total amount of green space cover	.81	***	←
Average patch size of bare soil	.38	**	←
Average patch size of buildings	.29	*	→
Average patch size of lawns	.10		
Average patch size of impervious surface	.26	*	→
Average patch size of tree cover	.58	***	←
Average patch size of green space	.67	***	←
Total basal area of trees in the transect	.09		
Average height of trees in the transect	.51	***	↘
Average DBH of trees in the transect	.10		
Percentage of trees with cavities in the transect	.37	***	←
Total assessed parcel value for property	.13		
Total gross parcel tax rate	.29	*	→
Mean parcel lotsize	.22	.	→
Mean gross floor area	.24	.	→
Mean living area of property	.26	*	→
Mean year property was built	.13		
Mean year property was last remodeled	.05		
Distance to green space ≥ 42 ha	.38	***	→

*** <.001, ** <.01, * <.05, . <.1.

unconstrained ordination that represents objects, in our case bird surveys, in a lower dimensional space, based on their dissimilarity. It is well suited to make the grouping structure of the objects visible and these structures can later be related to underlying ecological factors and gradients (Legendre & Legendre, 1998). Depending on the data type, there are several methods for calculating statistical dissimilarity. For species data, the measure has to account for double zeros, i.e., the abundance of rare species can be zero at many sites but that does not necessarily mean that these sites are similar (Legendre & Legendre, 1998). We selected the Bray–Curtis index that accounts for double-zeros and is suitable for abundance data, following the selection criteria provided by Legendre and Legendre (1998).

Preparing the data for calculating the dissimilarity, we first compiled a cross table (function *crossstab*) that contained the maximum number of individuals encountered during all observations in the two years for each site and each species. Species observed at only one site and one visit were omitted because they may have an unduly large influence on the dissimilarity measure (Legendre & Gallagher, 2001). Most birds were only recorded in abundances of one, while a few were very abundant (e.g., European Starling and House Sparrow). In order to deal with the large differences between the smallest and largest abundances that would distort the dissimilarity measure (Oksanen et al., 2012), data were double square root transformed. The NMDS was finally calculated using the function *metaMDS*, which also rotates the result so that the first dimension contains most variability. The goodness-of-fit measure between the original dissimilarity and the dissimilarity in reduced space is the so-called *stress* measure. Small values mean low stress and high goodness-of-fit (Legendre & Legendre, 1998).

Once the ordination was calculated, we explored how its structure was related to parameters described in Section 2.3. First, we plotted the three different site types to obtain a visual impression of their similarity. Then, we plotted foraging guilds, substrate types, and the percentage of introduced (non-native) species onto the graph. Environmental vectors that showed the strongest correlation with corresponding environmental parameters at the sites

were fitted using the function *envfit*. If, for example, a variable showed no gradient across the sites on the ordination, then it was not correlated with its configuration. If a variable showed a distinct gradient and was correlated with the ordination, then a vector that pointed towards highest observations was fitted. The significance of the correlations was calculated using a bootstrap method based on 1000 iterations.

2.4.2. Species richness

To analyze general differences in the bird richness amongst the site types, we compiled species accumulation curves according to Colwell, Mao, and Chang (2004), using function *specaccum*. We also tested for differences using the non-parametric Wilcoxon test (function *wilcox.test*). To gain a better understanding of what constitutes the differences between site-pairs, excluding the large parks, we subtracted the number of species found at the random urban sites from their respective greening project. Likewise, the respective differences of the environmental parameters between the site-pairs were calculated and compared to the species-richness difference in a correlation analysis. Finally, we selected the parameters that were significantly correlated ($p < .1$) with the difference in species richness for a linear model, in order to test how these factors explained possible differences in species richness. We tested the selected parameters for normal distribution (Shapiro–Wilk test of normality, function *shapiro.test*) and co-linearity, and removed all bivariate correlations that were above the threshold of $R^2 = .49$ (Pearson correlation). The final model was calculated with the function *lm*. The relative importance of the model parameters was extracted with an ANOVA (function *anova*).

3. Results

We recorded 43 species in two years of bird surveys: 28 in 2010 and 34 in 2011. We recorded 413 individuals in 2010 and 626 in 2011. After removing all shore birds, 36 species remained. Seventeen species were found exclusively at the large park sites and not at the other plots. A complete list of species is provided in the electronic supplement material.

3.1. Bird community

We chose a two dimensional solution for the NMDS, because its correlation with the original data was only slightly lower than for a three dimensional solution (Linear fit $R^2 = 0.95$ vs. 0.97 , calculated with the function *stressplot*), while being easier to interpret. Overall goodness-of-fit of the NMDS was low (*Stress* = .12). The greening projects and the random urban sites on the one hand and the large parks on the other, form two distinct groups along the first dimension (NMDS1), with the large parks represented on the left side of the ordination (Fig. 3). In addition to being distinct from the urban sites, the large parks also show strong variation amongst each other in the second dimension (NMDS2).

Several vectors of environmental parameters were significantly correlated with the NMDS configuration, falling in two main categories: parameters increasing towards the greening projects and the random urban sites, like impervious surface; and parameters increasing towards the large park sites, like tree cover. Only the vector for average tree height points towards the Stony Brook Reservation, sites with very large trees. The environmental vectors with a significance level $p < .001$ are plotted on the ordination. All correlation coefficients, significances and the general directions of all the environmental vectors on the ordination are summarized in Table 2.

In addition to the environmental vectors, Fig. 4A and B shows the percentage of impervious surface and tree canopy at each site. Both parameters differ strongly between the large parks and all other

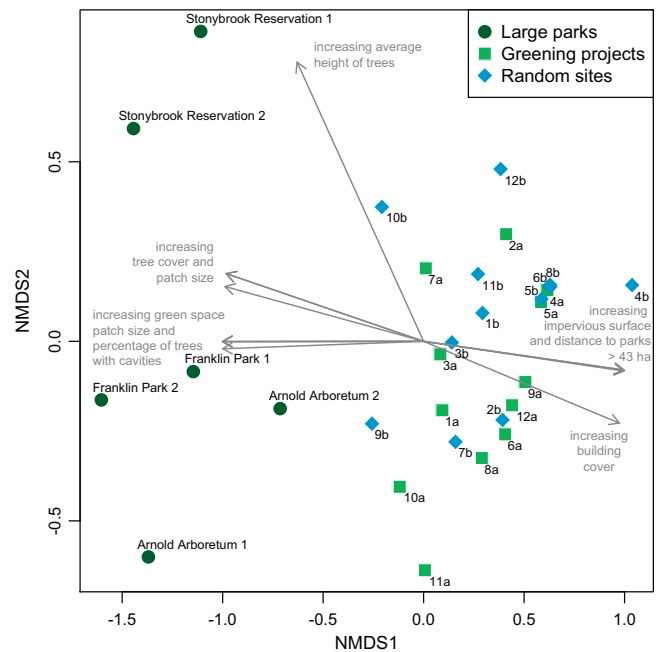


Fig. 3. Nonmetric multidimensional scaling (NMDS) ordination of the bird community at the 30 bird observation sites in Boston, Massachusetts. The NMDS represents sites according to their similarity in species composition and proximity in the statistical space indicates sites with a similar bird community. The map ID refers to Table 1 and Figs. 1 and 2. The arrows are vectors of environmental parameters that point towards where these parameters increase strongest. Only vectors with a significance $p < .001$ are shown (see also Table 2).

sites. In particular, the greening projects and random urban sites appear to be aligned along a gradient of impervious surface. Examining aspects of the bird community (Fig. 4C–F) further helps with the interpretation of the ordination. More insectivorous species (e.g., Tufted Titmouse, Downy Woodpecker, Scarlet Tanager) can be found towards the large parks with increasing tree canopy cover, one exception being the Chimney Swift, which forages in the open sky. The number of substrate types used for foraging shows a similar pattern. Both represent almost an inverse of the impervious surface at the sites. On the other hand, granivorous species increase with increasing impervious surface, as do introduced species.

3.2. Species diversity and richness

One-sided Wilcoxon tests showed that large parks had significantly higher bird richness than the greening projects ($p = .004$) and the random urban sites ($p = .001$) while the difference between the latter two showed low significance ($p = 0.06$) (Fig. 5A). The species accumulation curves revealed a similar pattern. Species richness at the large parks was much higher, despite the lower sample size, and the bands of the standard deviation do not overlap with greening projects and random urban sites. However, while the curve of the greening projects indicates that more species could be found at an equal number of random locations, the bands of the standard deviations overlap and the difference cannot be considered significant (Fig. 5B).

Both methods, the Wilcoxon tests and the species accumulation curves, do not allow for a direct comparison of site-pairs. However, using a paired one-sided Wilcoxon test, we found a difference between greening projects and random urban sites ($p = .049$). In fact, at eight out of the twelve site-pairs, the greening projects had more species than the random urban sites in their vicinity, two had the same amount and two had less (Fig. 6). Only four differences in

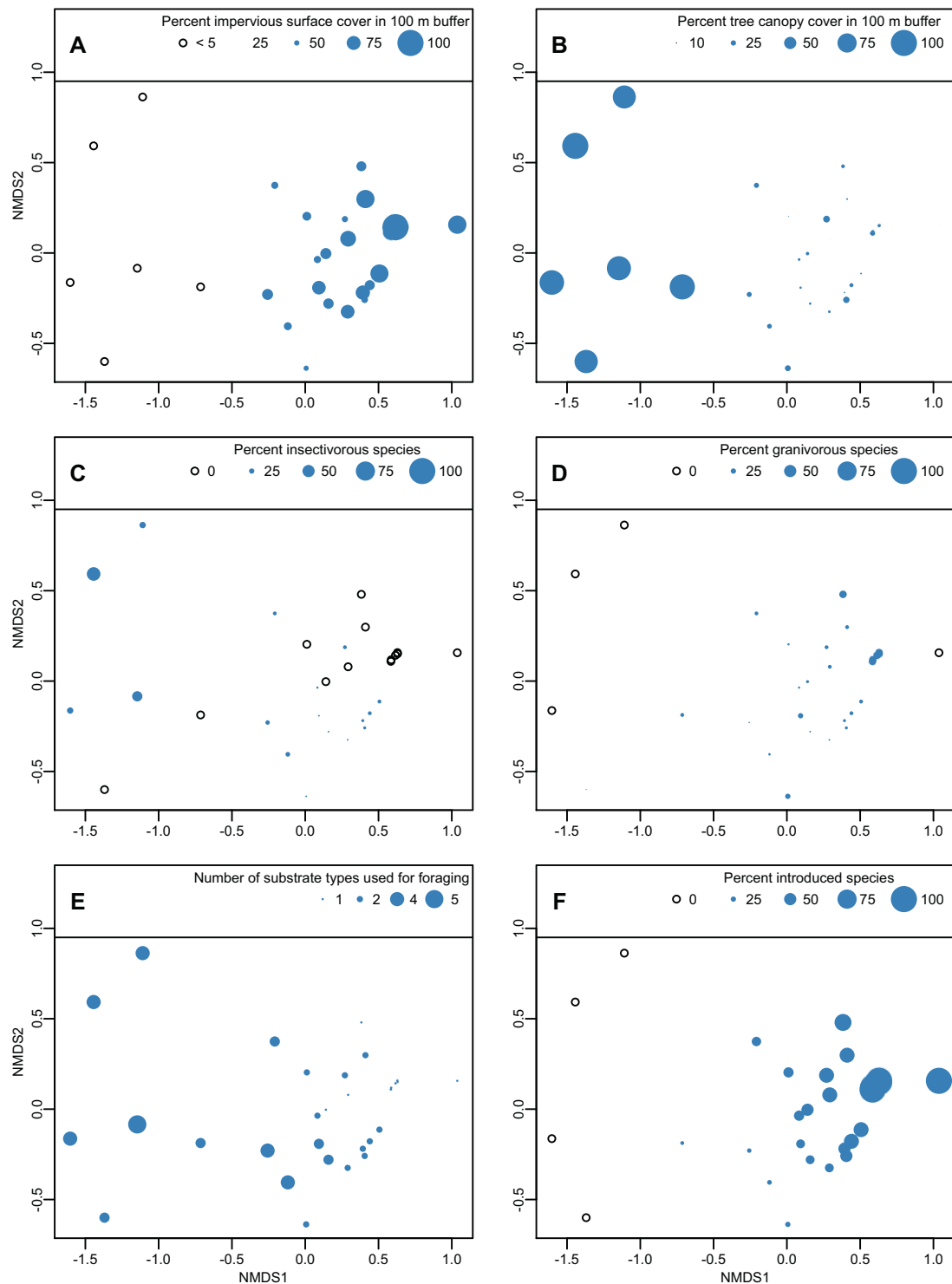


Fig. 4. The nonmetric multidimensional scaling (NMDS) ordination of the bird community at the 30 point count sites in Boston, Massachusetts, from Fig. 1 with an overlay of impervious surface (A), tree canopy cover (B), and bird community traits (C–F).

the environmental parameters were significantly correlated with the difference in species number (Table 3). Three are related to the amount of green space (tree canopy cover, total amount of green space cover, average patch size of green space) and one parameter is the percentage of trees with cavities in the transect.

For the linear model, we checked all four parameters for collinearity and normal distribution. Tree canopy cover, total amount of green space cover and the percentage of trees with cavities in

the transect were highly correlated with each other. According to the criteria described in Section 2.4, only the percentage of trees with cavities in the transect and the average patch size of green space were included in the regression and both were significant (see Table 4). Adding an interaction term did not improve the model. The overall explained variance by the model is $R^2 = .75$, the adjusted explained variance is $R^2 = 0.7$. The average patch size of green space explained 61%, the percentage of trees with cavities in the transect

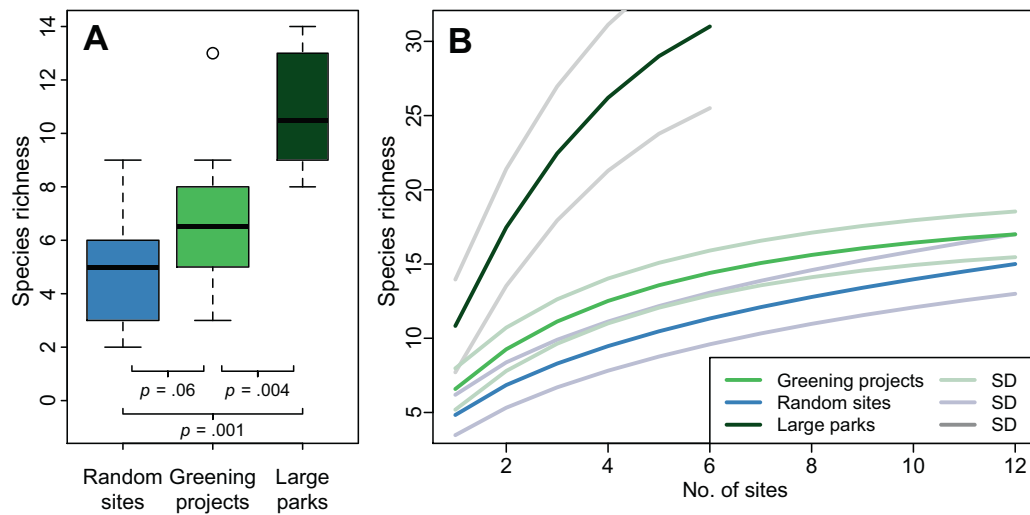


Fig. 5. Differences in species richness between greening projects, random urban sites and large parks. (A) Box-and-whisker plots of the species richness with significance levels p based on one-sided unpaired Wilcoxon tests. (B) Species accumulation curves with bands of standard deviation (SD).

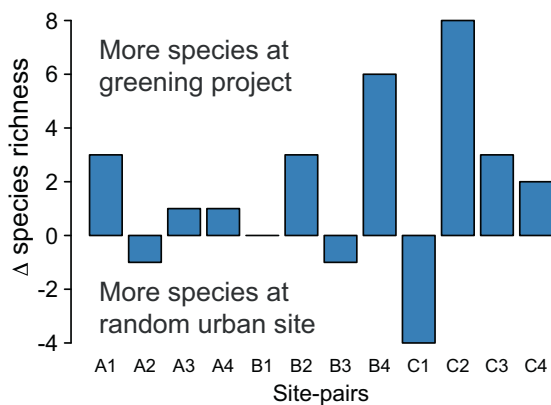


Fig. 6. The species richness differences of the site-pairs (greening projects – random urban sites). At most greening projects, more species were observed than at the random urban sites in their vicinity. The IDs refer to the sites in Table 1 and Figs. 1–3.

explained 15% of the total variance (Table 4). The model suggests that adding approximately 150 m² or 25% more trees with cavities leads to one more species being observed.

4. Discussion and conclusions

4.1. Bird community

Our results show that there are no general differences between the bird community composition of the greening projects and the random urban sites; they are mixed and do not form distinguishable groups on the ordination (Fig. 3). The impact of the greening projects is probably too small or the effect of the surrounding urban habitat is too strong to lead to detectable impacts in the community composition. There is, however, a clear distinction with the large

Table 3

Correlation between the difference in species richness and differences in environmental parameters for 12 site-pairs (greening project and random urban site). The table shows the correlation coefficient r and the significance p .

Parameter	r	p
Bare soil cover	-.27	
Building cover	-.24	
Lawn cover	.40	
Impervious surface cover	-.24	
Tree canopy cover	.53	
Total amount of green space cover	.55	
Average patch size of bare soil	-.28	
Average patch size of buildings	-.07	
Average patch size of lawns	.14	
Average patch size of impervious surface	.12	
Average patch size of tree cover	.47	
Average patch size of green space	.78	**
Total basal area of trees in the transect	-.43	
Average height of trees in the transect	-.39	
Average DBH of trees in the transect	-.31	
Percentage of trees with cavities in the transect	.51	
Total assessed parcel value for property	-.33	
Total gross parcel tax rate	.08	
Mean parcel lotsize	-.04	
Mean gross floor area	-.29	
Mean living area of property	-.24	
Mean year property was built	-.02	
Mean year property was last remodeled	.15	
Distance to green space ≥ 42 ha	-.02	

*** <.001, ** <.01, * <.05, · <.1.

parks, demonstrating their conservation value. These sites provide habitat for bird communities that mainly consist of native species, species that forage on a variety of substrates, and include specialist guilds like insectivores (Fig. 4C, E, and F) that have been shown to be sensitive to urbanization (White et al., 2005). The strong variation among the large park sites along the second dimension (NMDS2) is probably due to a sampling effect. While they are distinct from the

Table 4
Coefficients for linear model explaining the difference in species richness based on differences in environmental parameters at the 12 site-pairs (greening project and random urban site).

	Estimate	p value of Est.	df	Sum of squares	p value
Intercept	0.792	.181			
Average patch size of green space	0.007	.002	1	69.349	.001
Share of trees with cavities in the transect	4.092	.046	1	16.792	.045
Residuals			9	28.108	

other sites, they are so species rich that the four 5-min point counts over two years have probably not yet detected all species that are present. The species accumulation curves confirm that saturation has not been reached (Fig. 5B). In addition, there are probably habitat differences between the sites, as indicated by the large tree heights at Stony Brook Reservation (Fig. 3).

There are several environmental parameters for which significant vectors could be fitted onto the ordination result. All of them extend mainly along the first dimension of the ordination (NMDS1) and can be interpreted as a gradient of urbanization intensity, with environmental parameters such as distance to green space larger than 43 ha or impervious surface increasing along the first dimension of the ordination, and parameters like canopy cover, patch size of green space, and the percentage of trees with cavities decreasing along the first dimension. This is very much in accord with results from Bolger et al. (1997), Melles et al. (2003) and White et al. (2005). The increasing prevalence of introduced and granivorous species in highly urbanized areas also reflects results from the review by Chace and Walsh (2006).

4.2. Species richness

While the bird community composition does not appear to be different at the greening projects compared to the random urban sites, the paired test showed that a difference in species richness was moderately significant and eight greening projects had more species present than the random urban site in the vicinity (Fig. 6). In fact, McCaffrey and Mannan (2012) and Hostetler and Knowles-Yanez (2003) showed that different bird species respond to urban habitat feature at different scales, some of which can be very local (100 m radius). In our model, the difference in species richness was mainly explained by the difference in the patch size of green space and to a lesser extent by the percentage of trees with cavities. The large influence of patch size fits within the theory of island biogeography, a theory that explains species richness of islands by their size and their distance to larger islands, and that can also be applied to terrestrial habitats and to urban ecosystems (Fernández-Juricic & Jokimäki, 2001; MacArthur & Wilson, 1967). Jokimäki (1999) found that the total area was the most important factor for bird richness in parks between .1 and 7.0 ha size in Oulu, Finland (explaining 39% of the variance). Similar results were found for small parks (0.2–2.0 ha) in Pachuca, Mexico (Carbó-Ramírez & Zuria, 2011). Our model suggests that an increase of the average patch size of green space by 150 m² results in the observation of one more species. Most of the greening projects are larger than this.

In addition to patch size, the percentage of trees with cavities also explains some of the species richness differences in Boston; in our model, 15% of the total variance of species richness differences was explained by this parameter. The importance of trees with cavities for many urban birds was highlighted by Blewett and Marzluff (2005). Trees with cavities provide nesting opportunities for many species and should be preserved when they do not serve as a safety hazard. In many parts of cities, trees have to be pruned, but nesting boxes can provide alternative opportunities for some bird species in cities (Fernández-Juricic & Jokimäki, 2001; Snep & Opdam, 2010). It must be clarified that the difference in trees with cavities is unlikely to stem from the recent tree plantings but rather from trees that were present at and around the sites before the “CityRoots” projects were implemented. Therefore, this is an indicator of the tree maintenance level and tree age surrounding the sites. Nevertheless, the greening projects can serve to protect existing tree habitat.

4.3. Implications for conservation

Large green spaces act as important refuges for wildlife in cities (Donnelly & Marzluff, 2004; Fernández-Juricic & Jokimäki, 2001;

Sandström et al., 2006). This is also the case in Boston where large parks have much higher species richness, more specialist guilds and fewer introduced species than the small greening projects and the random urban sites (Figs. 4 and 5). However, in many cities, the majority of people live far away from nature reserves and other species rich areas (Turner, Nakamura, & Dinetti, 2004) and the interaction between humans and nature increasingly depends on non-formal green space (Fuller & Gaston, 2009). Our results suggest that visitors of such green space are likely to experience a few more species than in the surrounding urban area. This is important, considering that experiences with biodiversity has been linked to human well-being (Fuller, Irvine, Devine-Wright, Warren, & Gaston, 2007) and is seen as a prerequisite for engaging urban dwellers in conservation efforts (Miller & Hobbs, 2002).

Our model suggests that an additional 150 m² in green space patch size accounts for one additional species being observed. However, whether the greening projects just attract birds that are present in the area or whether they actually provide foraging or nesting habitat to support additional species cannot be determined with our study, but nest monitoring and experiments with artificial nests could show this in the future. Because our study does not provide a before and after comparison, it cannot be ruled out that species richness differences existed even before the implementation of the greening projects. Again, long term monitoring of before and after greening would be helpful.

Results from this and other studies indicate that patch area matters and new green space will have the highest benefit for conservation when it is large and close or connected to larger patches of existing green space (Andersson & Bodin, 2009; Donnelly & Marzluff, 2004; Fernández-Juricic & Jokimäki, 2001; Sandström et al., 2006). Large and connected green space represents a potential tradeoff with the goal of greening initiatives such as “CityRoots,” which specifically target underserved neighborhoods and rely on the local community for construction (Urban Ecology Institute, n.d.). Projects implemented with these objectives are often built in areas with little initial green space and are usually not bigger than one or two vacant parcels (Table 1 and Fig. 2). To the extent that funding for greening projects is limited, community leaders and planners may have to choose between prioritizing biodiversity and environmental justice goals.

Increasing the bird diversity in neighborhoods with little existing green space would require a green network, a classic example of which is the Emerald Necklace in Boston created by the visionary Olmsted (Swaffield, 2005). While Olmsted did not plan just for bird conservation, he was aware of the multiple functions and the great value of large and interconnected green space (McMahon, 2000). Future green networks have to include public and private green space alike (Barker, 1997; Swaffield, 2005). Residential yards and gardens can provide valuable foraging and nesting habitat for urban birds (Cannon, 1999; Chamberlain, Cannon, & Toms, 2004; Lerman & Warren, 2011) and cover large quantities of urban areas (Loram, Tratalos, Warren, & Gaston, 2007). National programs such as the National Wildlife Federation program to certify yards as wildlife habitat (<http://www.nwf.org/backyard/>) and the National Audubon Society “healthy yards” program (http://www.audubon.org/bird/at_home/Healthy_Yard.html) represent small greening efforts on private lands. As far as we are aware, the efficacy of these programs has yet to be tested with regard to factors that contribute to species fitness. When extending the concept of urban greening to include habitat resources, solitary bees and wasps responded positively to the inclusion of artificial nest sites in Sheffield, UK, domestic gardens (Gaston et al., 2005), suggesting the positive impact of such efforts towards increasing urban biodiversity. Additionally, a city-wide survey in Sheffield revealed that 14% of all households had ponds, 26% had nest boxes, 29% had compost heaps, and 48% had trees in their gardens

(Gaston et al., 2005). Whilst these spatially disparate greening efforts might not affect local diversity, they have the potential to scale up and improve regional diversity (Goddard, Dougill, & Benton, 2010). Green roofs and green walls provide additional foraging and nesting habitat for birds when well designed and managed (Chiquet, Dover, & Mitchell, 2012; Fernandez-Canero & Gonzalez-Redondo, 2010).

In a green network, small greening projects could be designed to provide key habitat features like fruit bearing trees, linear elements like hedges that provide habitat and improve permeability of the urban matrix, nesting boxes, and in the long-run, more trees with cavities (Fernández-Juricic & Jokimäki, 2001). In addition, if the habitat quality of the urban matrix (gardens, streetscapes, etc.) increases, then the importance of small parks might increase. Murgui (2009), who studied bird richness in Valencia, Spain, using a raster survey, found that the total area covered by urban parks was more important for explaining the number of bird species than mean patch size of parks. However, the result was confined to winter birds and the author suggested that general habitat enhancements in the city would be needed for increasing the number of breeding birds. Several small greening projects, like the ones studied in this paper, could probably increase the habitat value of an area. In combination with previous research, our findings suggest that greening initiatives like “CityRoots” or “Grow Boston Greener” would have to be integrated into a comprehensive green space plan to achieve the greatest value for bird conservation.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.landurbplan.2013.02.007>.

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